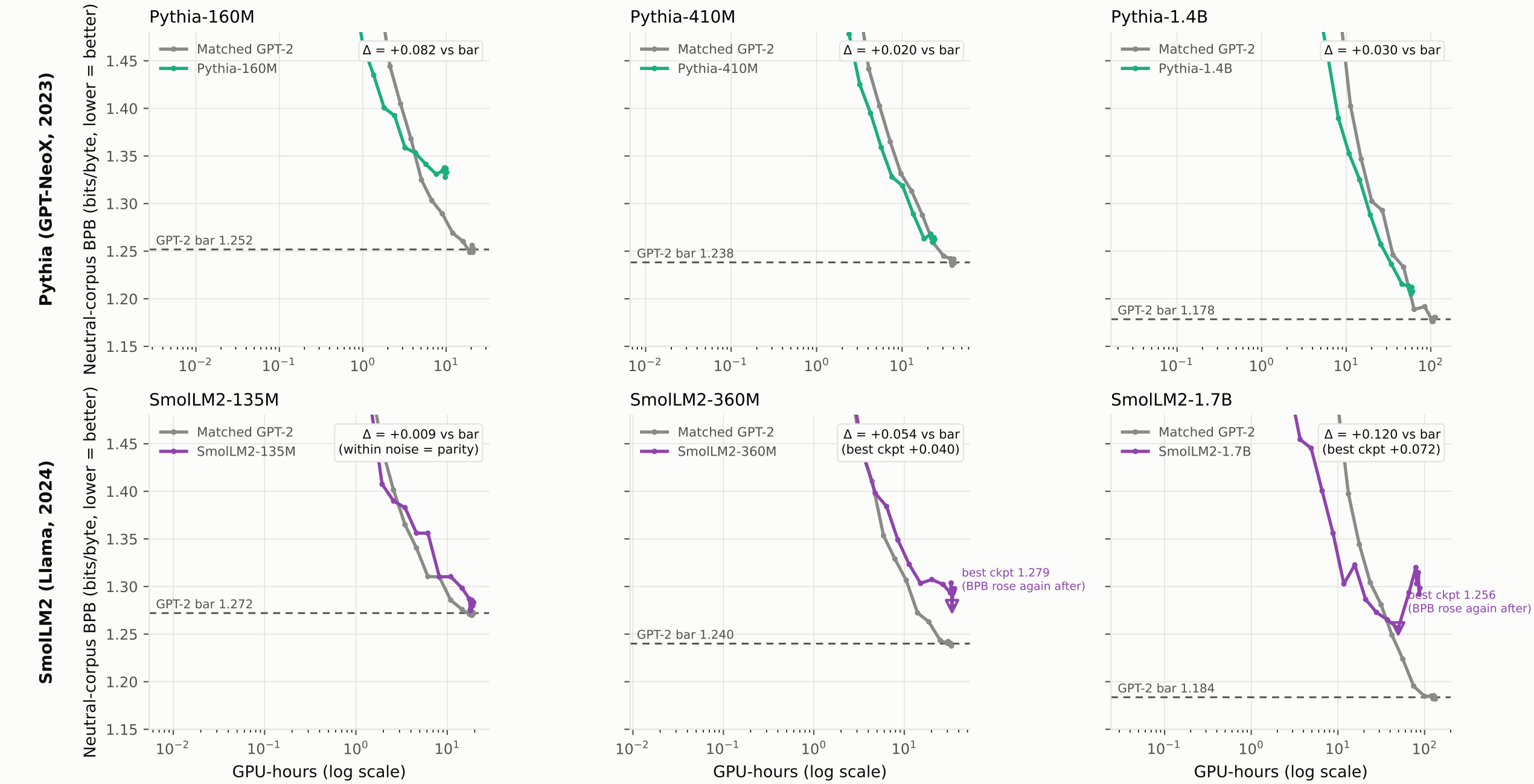


Bits per byte vs. GPU-hours for matched architecture comparisons



Each modern architecture vs a GPT-2 trained through the identical pipeline at the same size (OWT, 8.87B tokens, converged 512k-batch recipe). Lower BPB is better; Δ is the final BPB minus the matched-GPT-2 bar. Open triangle = best (lowest) checkpoint for SmoLM2-360M and SmoLM2-1.7B, whose BPB rose again later in training.